

Data I/O + Structure

Data Wrangling in R

Data Input

Outline

- ▶ Part 0: A little bit of set up!
- ▶ Part 1: reading in manually (point and click)
- ▶ Part 2: reading in directly & working directories
- ▶ Part 3: checking data & multiple file formats

Data Input: readr

`read_delim()` and `read_csv()` from the `readr` package

example for character delimited:

```
read_delim(file = "file.txt", delim = "\t")
```

comma delimited:

```
read_csv("file.csv")
```

Data Input

- ▶ The filename is the path to your file, in quotes
- ▶ The function will look in your “working directory” if no absolute file path is given
- ▶ Note that the filename can also be a path to a file on a website (e.g. ‘`www.someurl.com/table1.txt`’)

Example

https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_data_complete.csv

```
# From URL
ufo <- read_csv(
  "https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_data
)

# From your 'data-wrangling' directory
ufo <- read_csv("ufo_data_complete.csv")
```

(Warning message: One or more parsing issues, call
'problems()' – more on this later)

Data Input

The `read_delim()` and related functions return a “tibble” is a `data.frame` with special printing, which is the primary data format for most data cleaning and analyses.

```
class(ufo)
```

```
[1] "spec_tbl_df" "tbl_df"      "tbl"        "data.frame"
```

Check to make sure you see the new object in the Environment pane.

Data Input

There are also data importing functions provided in base R (rather than the `readr` package), like `read.delim` and `read.csv`.

These functions have slightly different syntax for reading in data, like `header` and `as.is`.

However, while many online resources use the base R tools, recent versions of RStudio switched to use these new `readr` data import tools, so we will use them here. They are also up to two times faster for reading in large datasets, and have a progress bar which is nice.

Data Input: readr

`read_table()` from the `readr` package, allows any number of whitespace characters between columns, and the lines can be of different lengths.

```
# example for whitespace delimited :  
read_table(file = "file.txt")
```

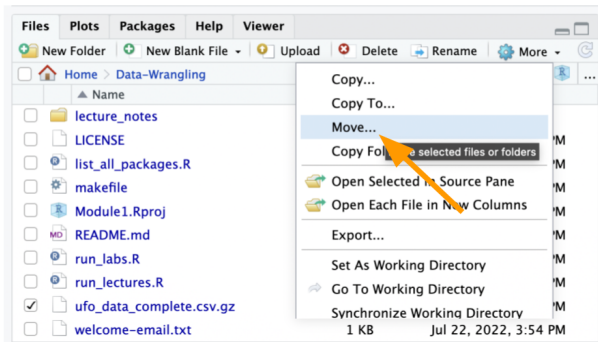
Clean the data while you read it in!

The argument `trim_ws` removes trailing and leading spaces around your data.

```
# example:  
read_csv(file = "file.txt", trim_ws = TRUE)
```

Data Input - working directories

What if your file is in the “Home” directory?



Data Input

Backtrack using the relative path with ../ like:

```
ufo <- read_csv("../ufo_data_complete.csv.gz")
```

Data Input

Or, read in from a subfolder:

```
ufo <- read_csv("data/ufo/ufo_data_complete.csv")
```

Warning: One or more parsing issues, call ``problems()`` on y
e.g.:

```
dat <- vroom(...)  
problems(dat)
```

Rows: 88875 Columns: 11

-- Column specification -----

Delimiter: ","

chr (10): datetime, city, state, country, shape, duration

dbl (1): duration (seconds)

i Use ``spec()`` to retrieve the full column specification for

i Specify the column types or set ``show_col_types = FALSE``

Check the data + other formats

Check the data out

- ▶ Some functions to look at a data frame:
 - ▶ `head()` shows first few rows
 - ▶ `tail()` shows the last few rows
 - ▶ `View()` shows the data as a spreadsheet
 - ▶ `spec()` gives specification of column types
 - ▶ `str()` gives the column types and specs
 - ▶ `glimpse()` similar to `str` (dplyr package)

What did I just read in?

- ▶ `nrow()` displays the number of rows of a data frame
- ▶ `ncol()` displays the number of columns
- ▶ `dim()` displays a vector of length 2: # rows, # columns

```
nrow(ufo)
```

```
[1] 88875
```

```
ncol(ufo)
```

```
[1] 11
```

```
dim(ufo)
```

```
[1] 88875    11
```


All Column Names

- ▶ `colnames()` displays the column names

```
colnames(ufo)
```

```
[1] "datetime"           "city"                 "state"
[4] "country"            "shape"                "duration"
[7] "duration (hours/min)" "comments"             "date posted"
[10] "latitude"           "longitude"
```

Column names and classes using `glimpse()`

```
glimpse(ufo)
```

Rows: 88,875

Columns: 11

[illegible]

Data Input

- ▶ Sometimes you get weird messages when reading in data.
- ▶ The `problems()` function shows you any issues with the data read-in.

```
head(problems(ufo))
```

```
# A tibble: 6 x 5
```

	row	col	expected	actual	file
	<int>	<int>	<chr>	<chr>	<chr>
1	878	12	11 columns	12 columns	/Users/avahoffman/Droph
2	1713	12	11 columns	12 columns	/Users/avahoffman/Droph
3	1815	12	11 columns	12 columns	/Users/avahoffman/Droph
4	2858	12	11 columns	12 columns	/Users/avahoffman/Droph
5	3734	12	11 columns	12 columns	/Users/avahoffman/Droph
6	4756	12	11 columns	12 columns	/Users/avahoffman/Droph

```
dim(problems(ufo))
```

```
[1] 199 5
```

Data input: other file types

- ▶ For reading Excel files, you can do one of:
 - ▶ use `read_excel()` function from `readxl` package
 - ▶ use other packages: `xlsx`, `openxlsx`
- ▶ `haven` package has functions to read SAS, SPSS, Stata formats

Selecting Excel sheets

Use the `sheet` argument to indicate which sheet to pull from. It can refer to the sheet's index or name.

example:

```
read_excel(path = "file.xlsx", sheet = 2)  
read_excel(path = "file.xlsx", sheet = "data")
```

After hours of cleaning. . . output!

Data Output

While its nice to be able to read in a variety of data formats, it's equally important to be able to output data somewhere.

`write_delim()`: Write a data frame to a delimited file

`write_csv()`: Write a data frame to a comma-delimited file

This is about twice as fast as `write.csv()`, and never writes row names.

Data Output

For example, we can write back out just the first 100 lines of the ufo dataset:

```
first_100 <- ufo[1:100,]  
write_delim(first_100, file = "ufo_first100.csv", delim = ",")  
write_csv(first_100, file = "ufo_first100.csv")
```


More ways to save: write_rds

If you want to save **one** object, you can use `readr::write_rds` to save to a compressed rds file:

```
write_rds(ufo, file = "ufo_dataset.rds", compress = "xz")
```

Read it back in:

```
ufo_new <- read_rds(file = "ufo_dataset.rds")
```

Tip: the `compress = "xz"` argument saves on file size!

More ways to save: save

The save command can save a set of R objects into an “R data file”, with the extension .rda or .RData.

```
x = 5  
save(ufo, x, file = "ufo_data.rda")
```

The opposite of save is load.

```
load(file = "ufo_data.rda")
```

Summary & Lab

- ▶ Use `read_delim()`, `read_csv()`, `read_table()` for common data types
- ▶ These have helpful `trim_ws` and `na` arguments!
- ▶ `read_excel()` has the `sheet` argument for reading from different sheets of the Excel file
- ▶ Many functions like `str()`, `View()`, and `glimpse()` can help you understand your data better
- ▶ Save your data with `write_delim()` and `write_csv()`

<https://sisbid.github.io/Data-Wrangling/labs/data-io-lab-part2.Rmd>